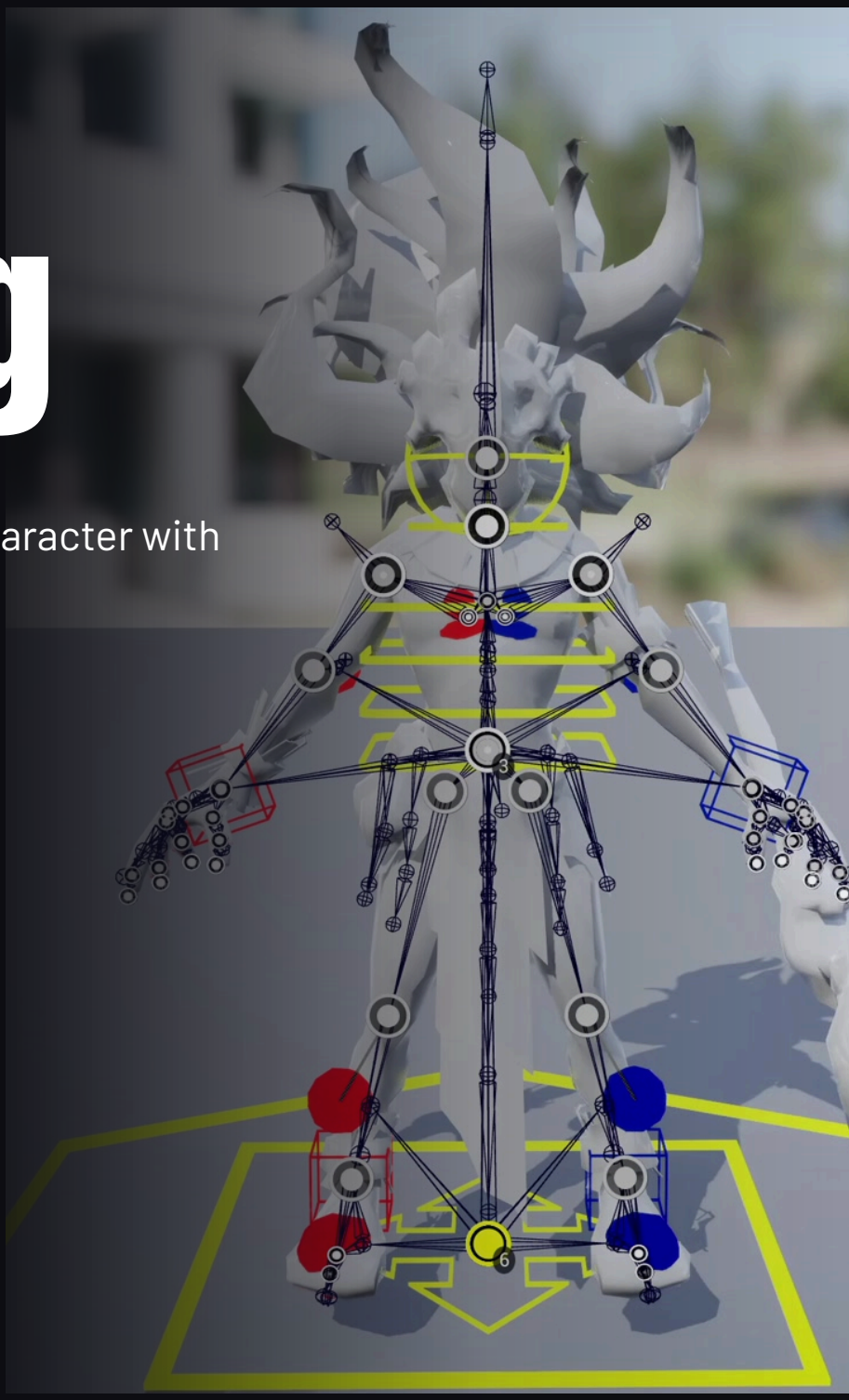
UNREAL ENGINE PLUGIN

AnyRig

Edit, rig and retarget any character with Control Rig.



USER GUIDE · VERSION 1.0 · UNREAL ENGINE 5.3 – 5.8

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1 What AnyRig does

AnyRig takes the slow, manual parts of character animation in Unreal Engine and does them for you, so you can spend your time animating.

TOOL	WHAT IT SOLVES
Animate	Edit any Animation Sequence with Control Rig right in the animation editor, with fewer keys to deal with (1 key per second instead of 30), Smooth for motion-capture glitches and one click to open the take in Sequencer.
Motion Solve	Move the character's body and AnyRig adds the foot steps, hip dip, chest lean and head follow-through.
Automatic Rig	Builds a full Control Rig for a character with any skeleton (body, neck, arms, legs, fingers, IK/FK) with Epic's Modular Rig, in one click.
Retarget Pro	Moves an animation between characters with different skeletons, even when Unreal's own retargeter cannot match them.
Bone equivalents	Shows, and lets you fix, which bone of your character is which bone of the UE5 Manny.
Compare	Explains why two skeletons do not share animations: names, hierarchy, rotations, proportions.
Animation Scaled	Fixes the skeleton's translation retargeting options in one click.

GOOD TO KNOW AnyRig never renames or modifies your original skeleton, mesh or animation. Edits are saved as new copies (or in Sequencer), and Epic's original Control Rig graphs are left untouched.

2 Installation

1. Download AnyRig from Fab, or unzip the package. It has one folder per engine version: `UE5.3`, `UE5.4` ... `UE5.8`.
2. Copy the `CharacterCompatibilityAnalyzer` folder for **your** engine version into your project's `Plugins` folder:
   `YourProject/Plugins/CharacterCompatibilityAnalyzer`
   Create the `Plugins` folder if it does not exist. (The folder keeps its technical name; in the editor the plugin shows as AnyRig.)
3. Open the project. The plugin is precompiled, so you do not need Visual Studio, and Blueprint-only projects work.
4. Open **Window › AnyRig**, or right-click any Animation Sequence.

Required Epic plugins

All of these ship with Unreal Engine. AnyRig enables them for you, and you can check them in **Edit › Plugins**:

- **Control Rig** and **IK Rig**: rigs and retargeting.
- **Sequencer Scripting**: opening and baking takes in Sequencer.
- **Control Rig Modules**: needed only for **Automatic Rig** (UE 5.4 and later).

USE THE MATCHING FOLDER Each engine version has its own build. A UE 5.6 build will not load in UE 5.7. If the editor says the plugin was built for another version, copy the folder for your exact version.

What the plugin adds to your project

- **Plugin content** (read-only reference): the UE5 Manny, its skeleton, its Control Rig and a pink material that marks AnyRig's reference assets.
- `/Game/CCA/`: what AnyRig creates for you, such as Automatic Rigs (`…_AutoRig`), retargeted animations, edit sessions and retarget setups. Delete anything you no longer need; AnyRig recreates it when needed.
- `Config/DefaultCharacterCompatibilityAnalyzer.ini`: your Edits (settings).

3 Unreal Engine versions

AnyRig supports UE 5.3 to 5.8. Epic changed several animation APIs between these versions, so a few features behave slightly differently. The tables below list every difference that affects you.

Feature availability

FEATURE	5.3	5.4	5.5	5.6	5.7	5.8
Animate (Control Rig in the animation editor)	✓	✓	✓	✓	✓	✓
Simplify, Smooth, keys per second	✓	✓	✓	✓	✓	✓
Motion Solve (animation editor and Sequencer)	✓	✓	✓	✓	✓	✓
Automatic Rig (Epic's Modular Rig)	–	✓	✓	✓	✓	✓
Retarget Pro, Bone equivalents, Compare	✓	✓	✓	✓	✓	✓
Bone recognition (names, languages, shape)	✓	✓	✓	✓	✓	✓

UE 5.1 and 5.2 are not supported.

Differences you may notice

AREA	VERSIONS	WHAT HAPPENS
Automatic Rig	5.3	Not available: Epic's Modular Rig, which it builds on, only exists from 5.4 on. Everything else works; use **Rig › Find Rig** with an existing Control Rig.
Automatic Rig: elbow and knee poles	5.4 – 5.5	Works. The module tree is addressed differently than in 5.6+, which AnyRig handles for you. Pole fine-tuning works on all supported versions.
Retargeting engine	5.3 – 5.5 5.6 +	5.3–5.5 use Epic's classic retarget processor; 5.6 and later use Epic's new retarget operation stack. Results are equivalent; on 5.6+ you can also add your own retarget operations in the IK Retargeter.
Epic's automatic bone alignment	5.3	Not in the engine. **Align to source** (feet, hands, head, legs) still works, using AnyRig's own alignment.
Hiding the original animation while you edit in Sequencer	5.3 – 5.5 5.6 +	On 5.6+ the original animation track is muted **only in your editor**. On 5.3–5.5 the track is muted in the sequence itself. Unmute it before rendering if you want both tracks.
Keys added per control in Sequencer	5.3 – 5.4	Keys go on the rig track's single section instead of one section per control. Same result in the animation.
Interpolation of solved keys	5.3 – 5.6 5.7 +	Motion Solve writes linear keys on 5.7+; earlier versions use Sequencer's default (auto) interpolation, so curves can look slightly softer. Simplify afterwards works the same.
Tween slider	5.3 – 5.5	The viewport may update only when you release the slider. On 5.6+ it updates live while you drag.
Bake a Sequencer take to an animation	5.3 – 5.5	Uses Sequencer Tools' exporter (needs the Sequencer Scripting plugin); 5.6+ use Control Rig's exporter. Same output.
Curves and sockets on the Manny rig copy	5.3 – 5.5	When AnyRig adapts the Manny Control Rig to your character, older engines copy the bones only; 5.6+ also copy curves and sockets. Automatic Rig copies curves from 5.5 on.
Control shapes	5.8	5.8 changed how control shapes are stored. AnyRig supports both formats; nothing to do on your side.

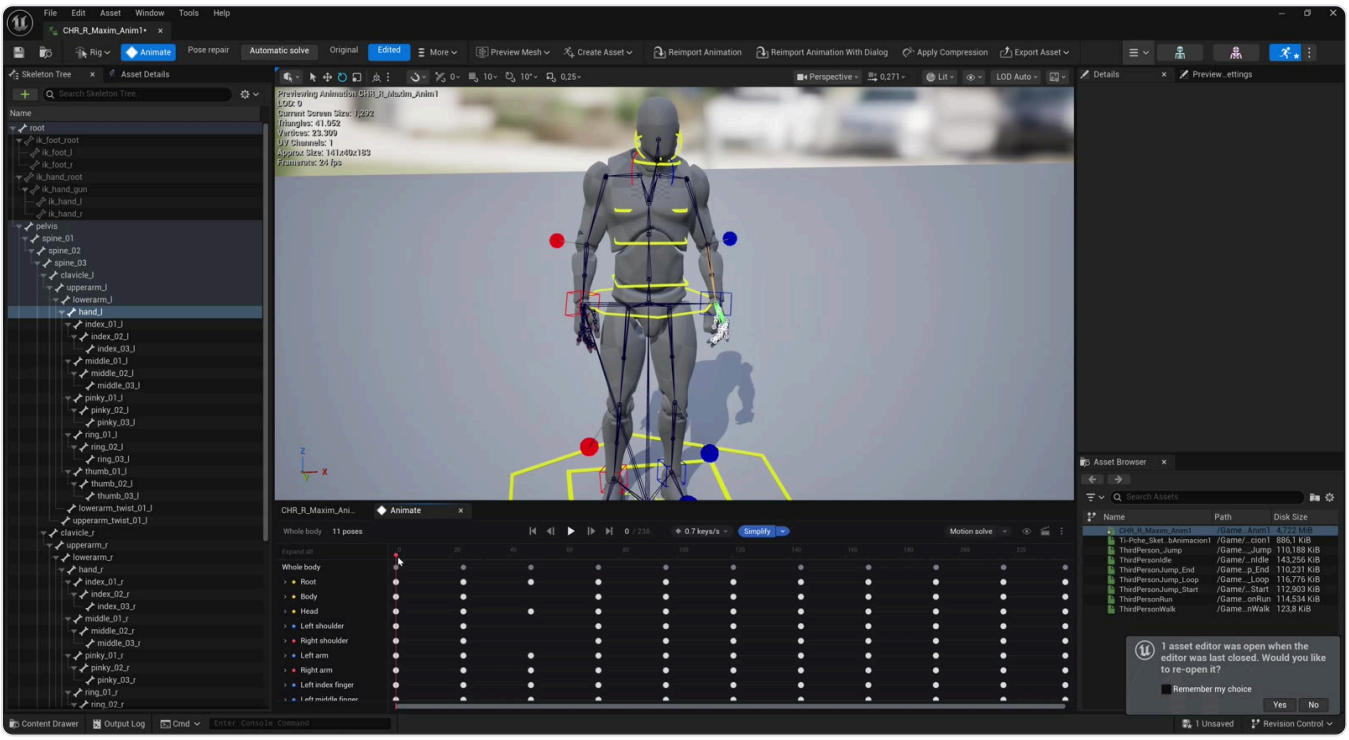
How each version was tested

VERSION	TESTING
5.7	Reference version: the full automated suite (33 tests: animate, keys, Motion Solve, rigs, retarget, compare, UI) on UE5, UE4, Mixamo, DAZ, Character Creator/iClone, MetaHuman and custom skeletons.
5.4, 5.5, 5.6, 5.8	Automatic Rig on five skeleton families (controls created and moving the right bones) and bone recognition on nine naming conventions.
5.3	Plugin loads, bone recognition on nine naming conventions.

4 Quick start

Edit an animation with Control Rig in under a minute.

1. Open any **Animation Sequence**.
2. In the toolbar, open **Rig** ▾. AnyRig lists the Control Rigs in your project that fit this character. Pick one, or choose **Automatic Rig** to build one (UE 5.4+).
3. Click **Animate**. The Animate tab opens under the viewport with the take converted to Control Rig keys.
4. Set **keys/s** to **1.0** for a clean timeline, then move controls in the viewport. Keys are added for you (Auto key).
5. Compare with **Original | Edited** in the toolbar, then save with **More** ▾ **› Bake copy**. Your original animation is never overwritten.



The Animate tab: body parts on the left, simplified keys (1 key per second) on the timeline, Control Rig controls in the viewport.

5 Animate

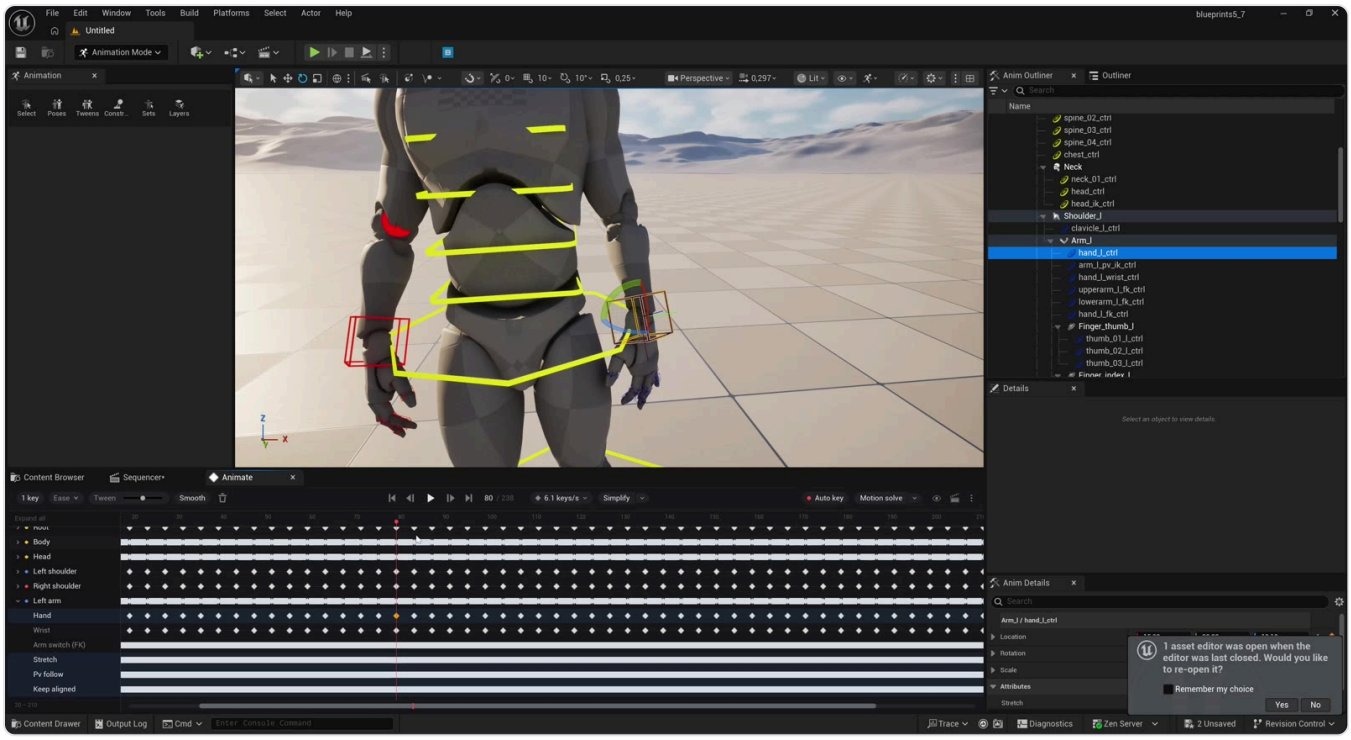
Control Rig inside Unreal's animation editor: no level, no Level Sequence and no extra assets needed.

Toolbar of the animation editor

BUTTON	WHAT IT DOES
Rig ▾	**Find Rig** (rigs in the project that fit), **Automatic Rig**, character setup, attach controls to bones, orientation guide, reset handle layout, IK/FK per limb, disable rig.
Animate	Opens the Animate tab with the take on Control Rig.
Pose repair	Moves control handles to where you want them without changing the rig graph or the animation.
Original | Edited	Switches the preview between your edit and the source animation.
More ▾	Motion Solve, edited keys, **Bake copy** (a new Animation Sequence), **Open in Sequencer**, preview in the level.

The Animate tab

- **keys/s** ▾ : show the take at 1, 2, 6, 12, 24, 30 or 60 keys per second, or every frame. Fewer keys means fewer things to fix; the motion between keys is interpolated. **Restore original keys** brings back the dense take.
- **Simplify** (on/off, with Precise / Balanced / Light): removes keys that do not change the motion within a tolerance. Turning it off returns to the original keys exactly and keeps your edits.
- **Smooth**: select keys and click Smooth to pull them toward their neighbours. One key smooths a spike; a range smooths a whole jittery stretch; click again for more. Ideal for motion-capture glitches.
- **Ease, delete, drag**: click or box-select keys (the box keeps growing while you scroll), drag them in time, change interpolation, delete.
- **Body parts**: controls are grouped (Root, Body, Head, shoulders, arms, legs, fingers). Clicking a control in the viewport opens its group. `Ctrl` +wheel zooms the timeline, `Shift` +wheel scrolls it, `Ctrl` + `A` selects all keys.
- **Auto key** (red dot): moving a control on a frame without a key adds one.



Smooth on a hand that jitters in a motion-capture take.

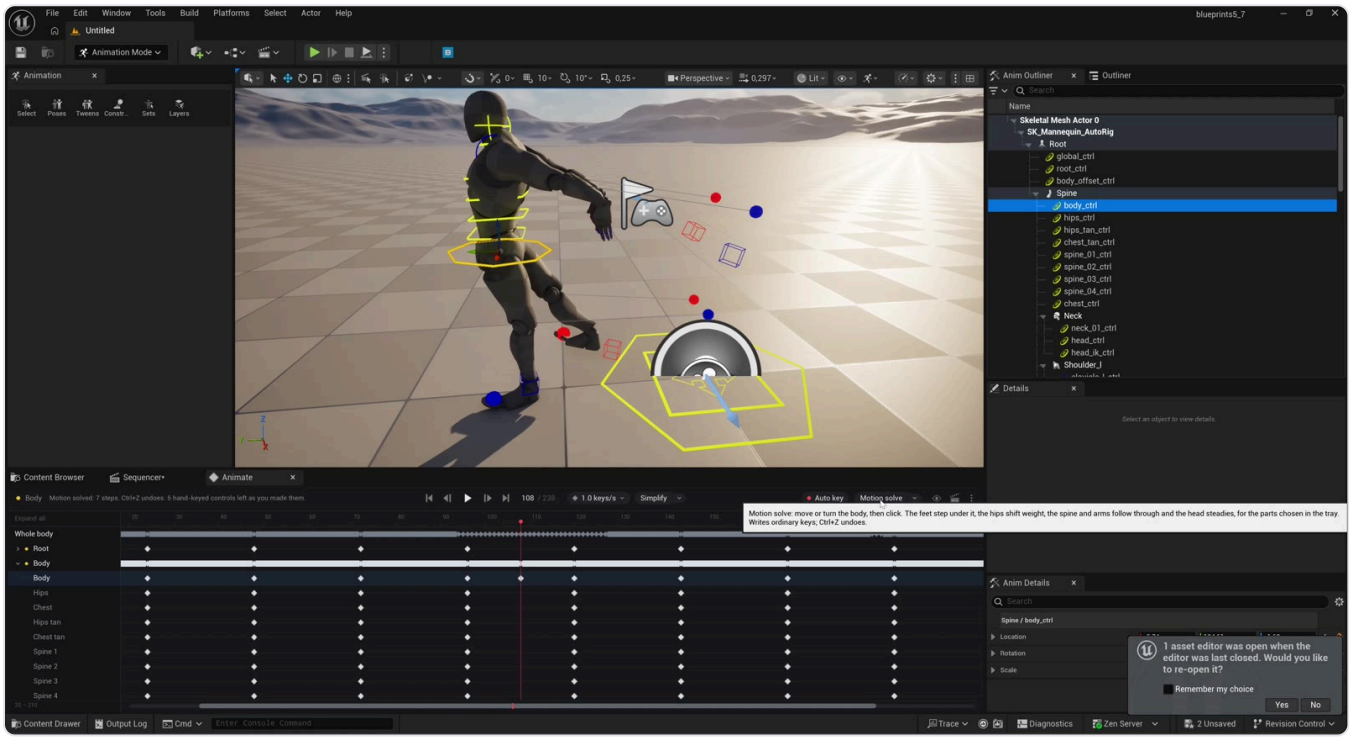
One click to Sequencer

More › Open in Sequencer builds a Level Sequence with your character and its Control Rig track, ready to animate, without dragging anything into the level. The same Animate tab (keys/s, Simplify, Smooth, Motion Solve, Auto key) is available in the level's Sequencer.

6 Motion Solve

Move the body; AnyRig adds what the rest of the body would do.

1. In the Animate tab, move or rotate the **Body** control over time, for example sliding the character 30 cm to the right.
2. Click **Motion solve**. The arrow next to it picks the parts: whole body, feet only, or feet, hips, spine, head and arms one by one.



Motion Solve stepping the feet after the body was moved.

- **Feet**: steps one foot at a time when the body moves or turns away from it, with lift, timing and an extra settling step when the body stops. Legs never over-stretch.
- **Hips** dip and shift toward the standing foot; **chest** leans against acceleration and swings back; **head** stays level and looks into turns.
- Keys you set by hand on a control are kept: the solve only replaces what it wrote itself. Running it again gives the same result; Ctrl+Z undoes it.
- Switching a limb between IK and FK runs the solve again for that limb.

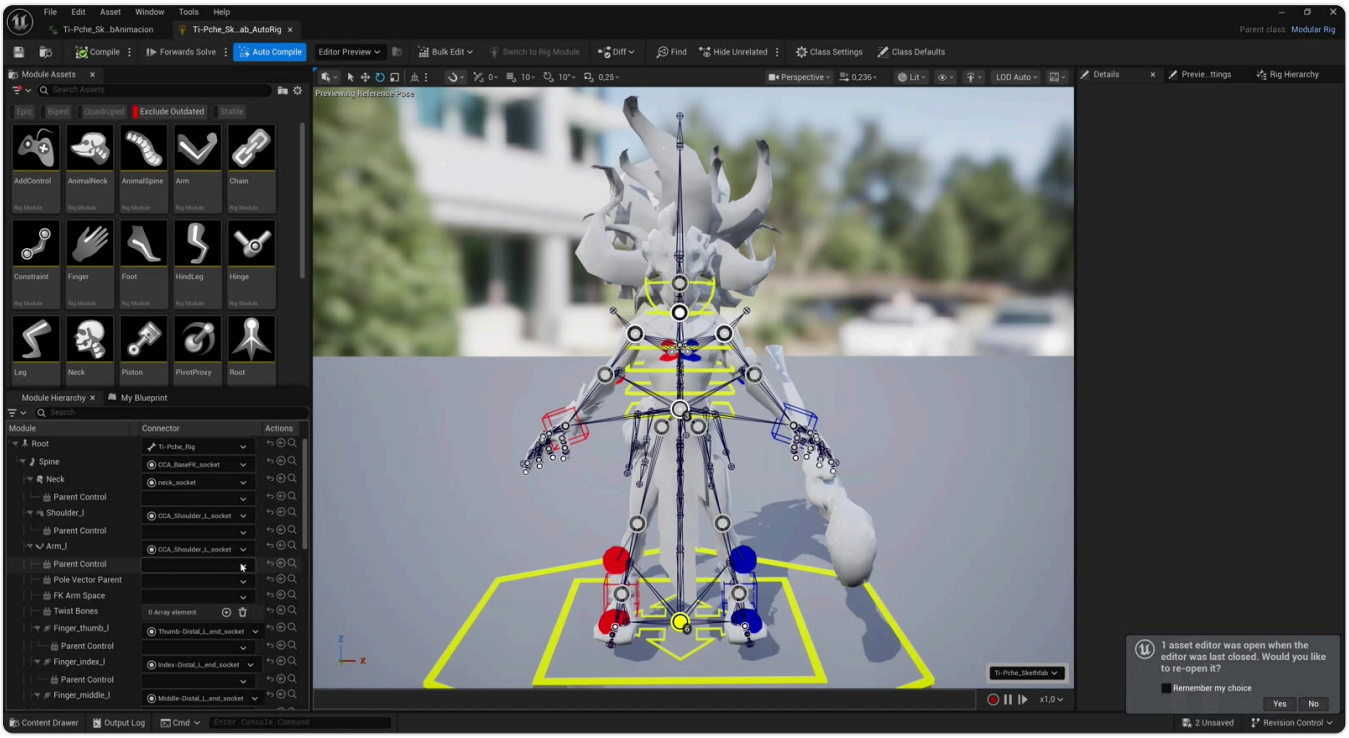
HANDS ARE UP TO YOU Motion Solve moves the feet and hips; it does not place hands on objects, because every animation needs them somewhere else. Arms only follow the body.

IN SEQUENCER The legs must be in IK for steps to appear (the solve does not change the take's IK/FK switches).

All timing and distances (step reach, step duration, lift, hip dip, chest lean, follow-through) are in **Edits › Motion solve**.

7 Automatic Rig UE 5.4+

A complete Control Rig for a character whose skeleton Unreal does not know, in one click.



A character from Sketchfab with a made-up skeleton, rigged by Automatic Rig in Epic's Modular Rig editor.

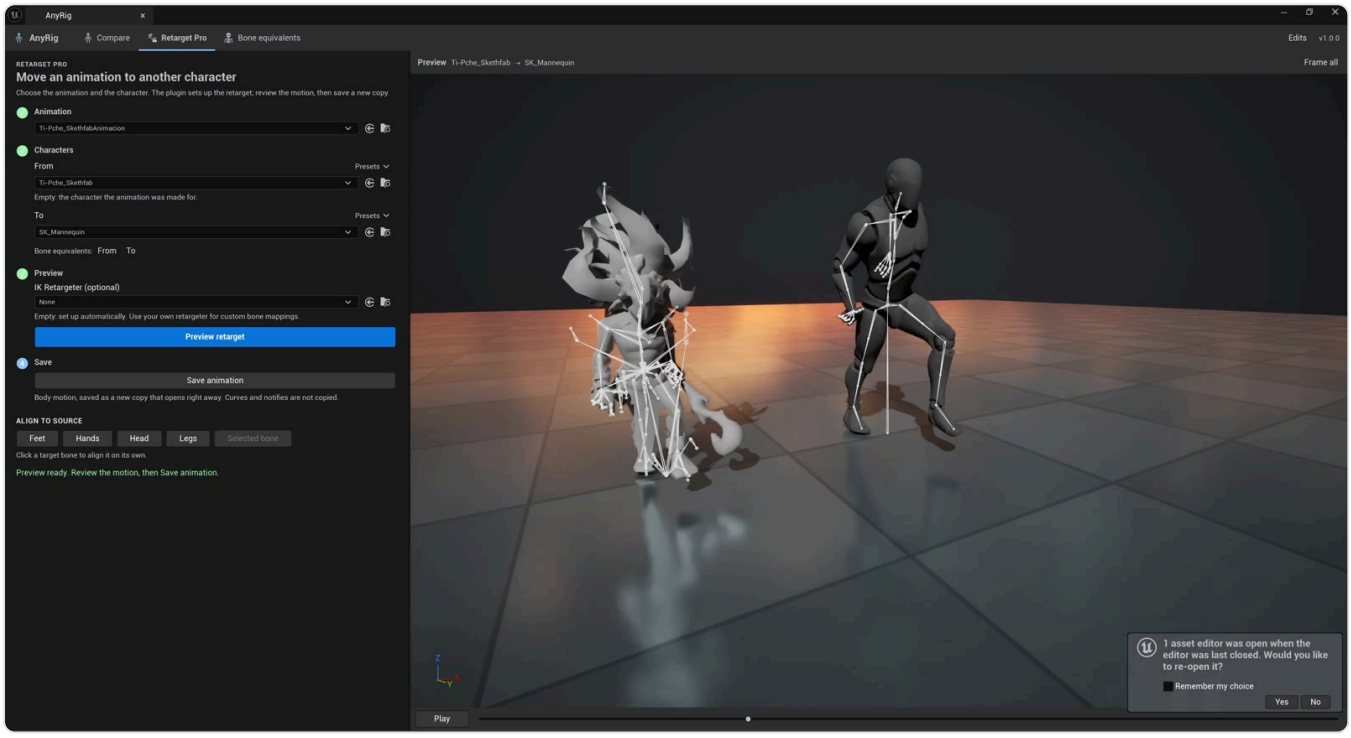
- Choose **Rig › Automatic Rig** in the animation editor, or **Create Rig** in AnyRig.
- AnyRig recognizes the body parts (see section 9) and builds Epic's Modular Rig modules: root, spine, neck, shoulders, arms, legs, feet and fingers, with IK/FK and pole vectors placed for this character.
- The rig is saved as `/Game/CCA/Rigs/<Mesh>_AutoRig` and opens in Epic's Modular Rig editor, where you can change any module.
- Rigs with a hip and spine as siblings (Blender style), huge scale (×100 exports) and characters without fingers or neck are handled.

ALWAYS REVIEW THE RIG Most skeletons work out of the box. Unusual ones (creatures, extra limbs) may need a small fix in the Modular Rig editor or in Bone equivalents.

8 Retarget Pro

Move an animation from one character to another, any skeleton to any skeleton.

1. **Animation**: pick the animation to move.
2. **Characters**: **From** (filled in from the animation) and **To**. Presets are shortcuts; any Skeletal Mesh works.
3. **Preview**: AnyRig builds the IK Rigs and the IK Retargeter for you and plays the result. You can also choose your own IK Retargeter.
4. **Save**: writes a new animation to the output folder (Edits › Output, default `/Game/CCA/Animations`).



A made-up skeleton (left) retargeted to the UE5 Manny (right). Unreal's own retargeter fails on this pair without manual setup.

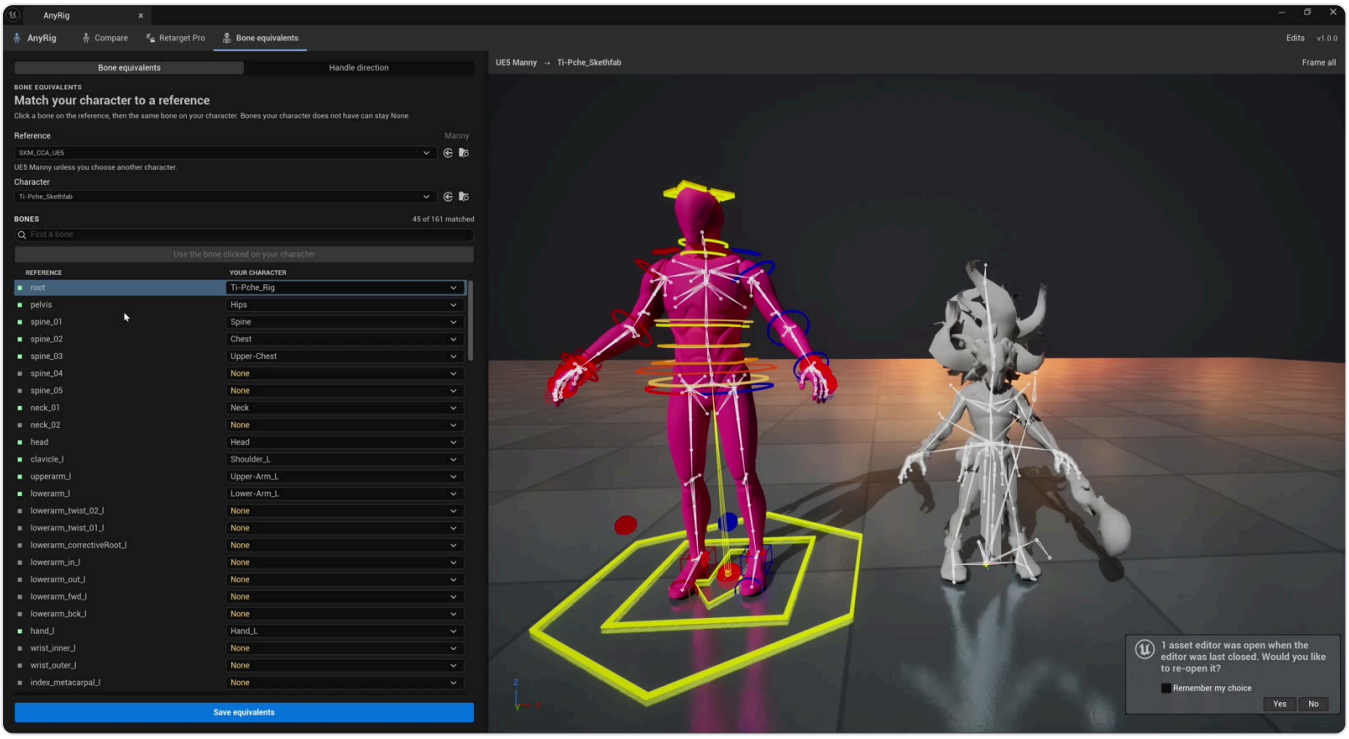
Fixing the result

- **Align to source**: feet, hands, head, legs or the selected bone match the source pose.
- Click a bone and rotate, move or scale it (snaps of 1/5/15°, 1 cm, 0.1); reset per bone or for all.
- Feet that turn inward are corrected; IK contacts follow your corrections.

LIMITS Curves and notifies are not all copied to the new animation yet. Additive animations need their base pose.

9 Bone equivalents and recognition

Everything in AnyRig starts by knowing which bone is the hip, the hand or the left thumb, whatever the bones are called.



Bone equivalents: the UE5 Manny on the left, your character on the right, matched bone by bone.

How bones are recognized

1. **Saved equivalents** you made in the Bone equivalents tab always win.
2. **Exact names** from a large table: Unreal (UE4, UE5, MetaHuman), Mixamo, DAZ, Character Creator/iClone, 3ds Max Biped, Unity/VRM, Maya, Blender/Rigify and MMD. Sides can be written many ways (L/R, Left/Right, Esq/Dir, Izq/Der, Gauche/Droite, Links/Rechts, Sx/Dx, 左/右...).
3. **Words in the name**, in English, Portuguese, Spanish, French, German, Italian, Japanese and Chinese, with typos (`Pelvs`, `Sholder`), dropped letters (`Hed`) and abbreviations (`Hnd`, `Thmb`). Exporter prefixes are ignored (`mixamorig:`, `Bip01`, `CC_Base_`, `DEF-`, `J_Bip_`...).
4. **The skeleton's shape**, when names say nothing: hands are where finger chains start, feet are the lowest limb ends, the chest is where both arms meet, the hips where legs and spine meet; sides come from the body's orientation.

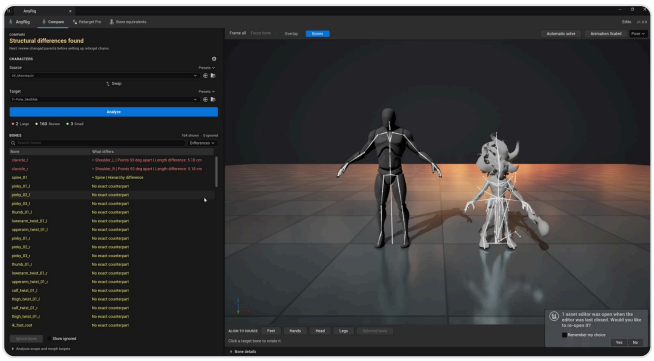
Making a correction

Open **Bone equivalents**, choose your character, click a Manny bone and then the bone of your character (in the list or the viewport), and **Save equivalents**. A bone can drive only one Manny bone, and parents must stay above children.

10 Compare and Animation Scaled

Compare

Pick a source (for example the Manny) and a target. AnyRig lists what differs, colored by size: **Large**, **Review** and **Small**. That covers names, missing or extra bones, parents, rotations, lengths and positions, plus LODs, materials, physics assets and facial bones.

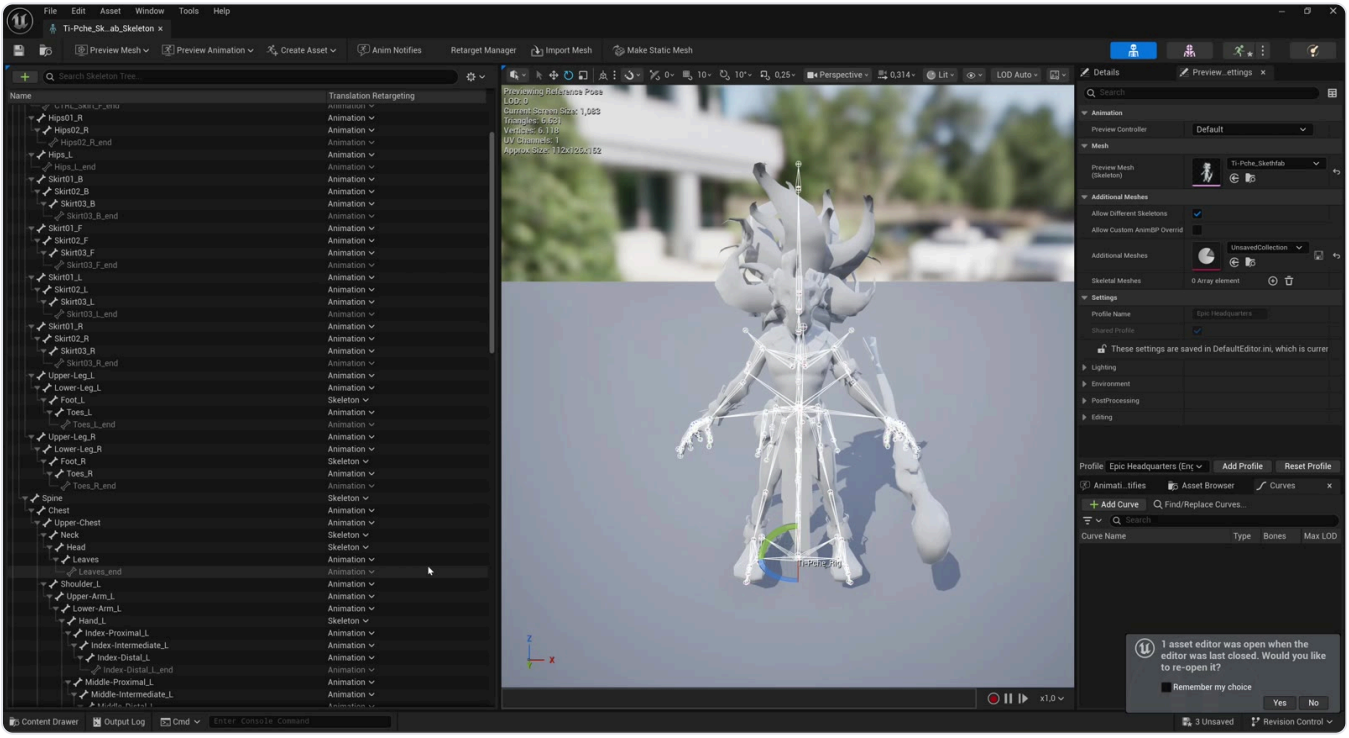


Compare listing the structural differences between two skeletons.

- Click a bone to focus it in the side-by-side or overlay preview.
- **Ignore bone** hides a difference you accept; it is remembered per project.
- Thresholds are in Edits › Compare.

Animation Scaled

Retargeted animations that stretch or shrink limbs usually come from a skeleton whose bones are set to *Animation* translation retargeting. That option is hidden in the Skeleton editor (**Show Retargeting Options**). **Animation Scaled** sets the right mode for every bone in one click.

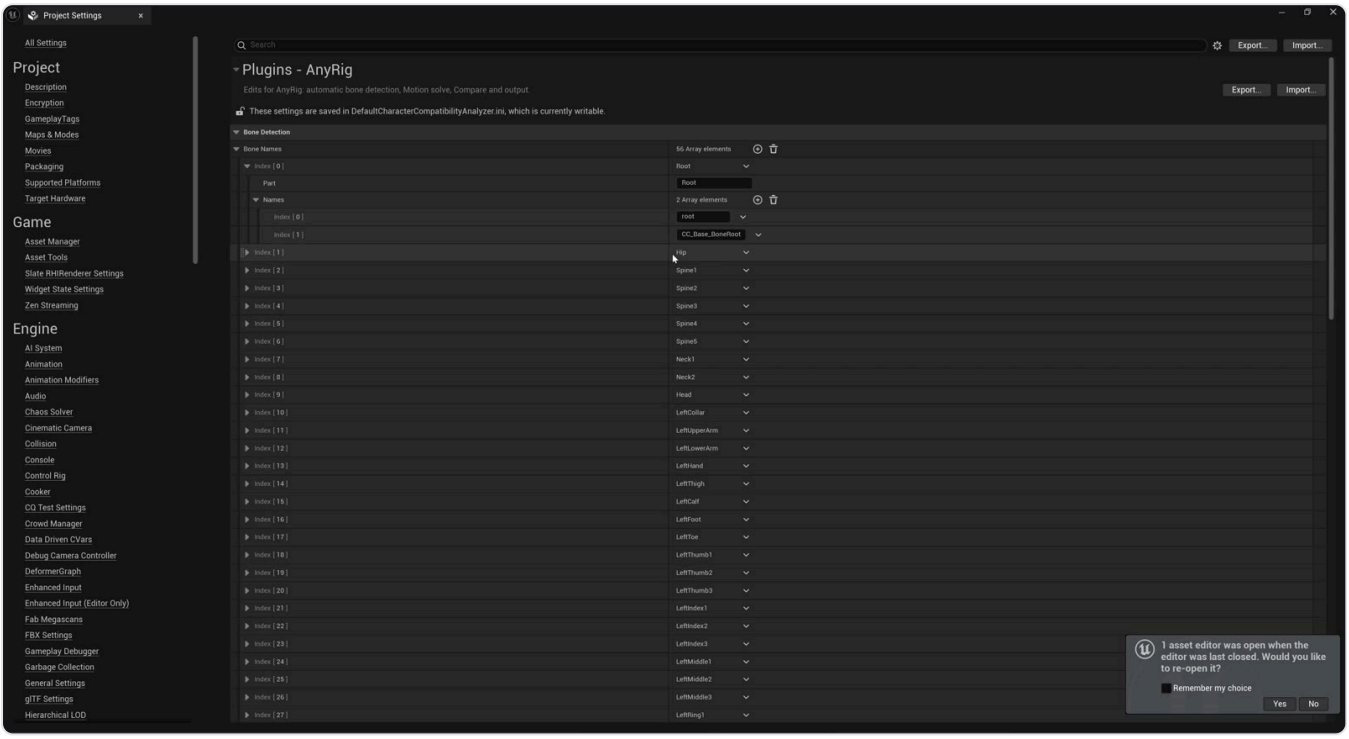


Translation retargeting options on a skeleton.

SAVE THE SKELETON Animation Scaled changes the Skeleton asset's retargeting options. Save the skeleton to keep them.

11 Edits (plugin settings)

Project Settings › Plugins › AnyRig, or the **Edits** button in the AnyRig window. Saved per project in `Config/DefaultCharacterCompatibilityAnalyzer.ini`; every value has a tooltip and a reset arrow.



The Edits page.

SECTION	SETTINGS
Bone detection	**Bone Names**: the body part → accepted bone names table; add your studio's names (e.g. a root bone called `dinosaur`). **Bone Words**: words per body part and side, in any language. **Smart detection**: words, typos and skeleton shape. Turn it off if you get wrong matches; exact names still work.
Motion solve	Step reach (with min/max), step duration, quickest step, step lift, hip dip, weight shift, chest lean, follow-through frequency and damping, head follow and look-ahead, Simplify tolerances after the solve.
Compare	Review/Large thresholds for rotation, proportion, position and scale.
Output	Folder where Retarget Pro saves animations.

12 Troubleshooting and FAQ

PROBLEM	WHAT TO DO
The plugin does not load / "built for another version"	Copy the folder that matches your exact engine version (section 2).
Automatic Rig is missing	You are on UE 5.3 (needs 5.4+), or **Control Rig Modules** is disabled in Edit › Plugins.
A bone is matched to the wrong part	Fix it in **Bone equivalents** and save; saved equivalents always win. Many wrong matches: turn off Smart detection in Edits.
Retargeted limbs stretch or shrink	Use **Animation Scaled** on the target skeleton and save it.
No steps after Motion Solve in Sequencer	Put the legs in IK, then solve again.
The original animation still plays under my edit in Sequencer (5.3–5.5)	The track is muted in the sequence on these versions. Check the mute state of the animation track.
Too many keys after editing	Turn on **Simplify** or lower **keys/s**.
Where are the assets AnyRig made?	`/Game/CCA/`. You can delete them; AnyRig recreates what it needs.

Does AnyRig change my files?

No. Originals are never renamed or overwritten. The one exception is Animation Scaled, which you click on purpose to change a skeleton's retargeting options.

Support

Questions, bugs and requests: open an issue at **github.com/gusvero/AnyRig/issues**, or use the Questions tab on the Fab listing. Please include your engine version, the character's skeleton (or a screenshot of its bone tree) and the Output Log.